

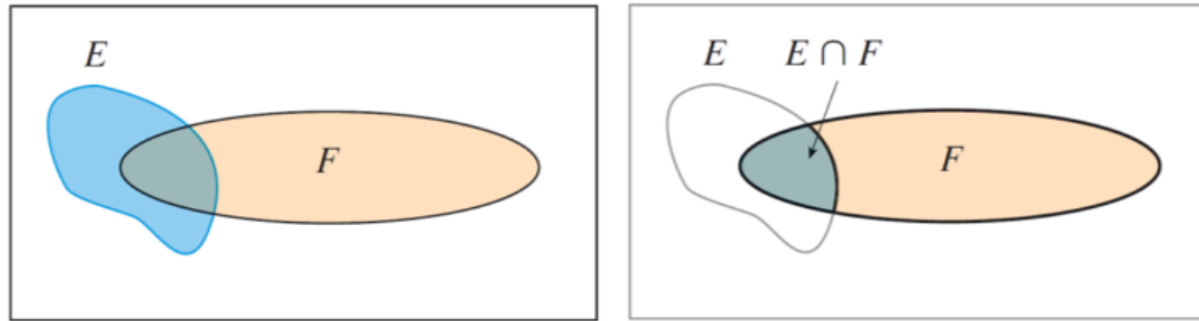
Likelihood in Phylogenetics

*What matters in probability is almost never
one single probability field, but rather
interrelatedness of many probability fields
(Freudenthal 1973, pp. 613).*

Background Knowledge

- Conditional probability
- Total probability
- Phylogenetic tree notation (Newick/Computer)

Conditional probability



Definition of **conditional probability** or Bayes' Theorem

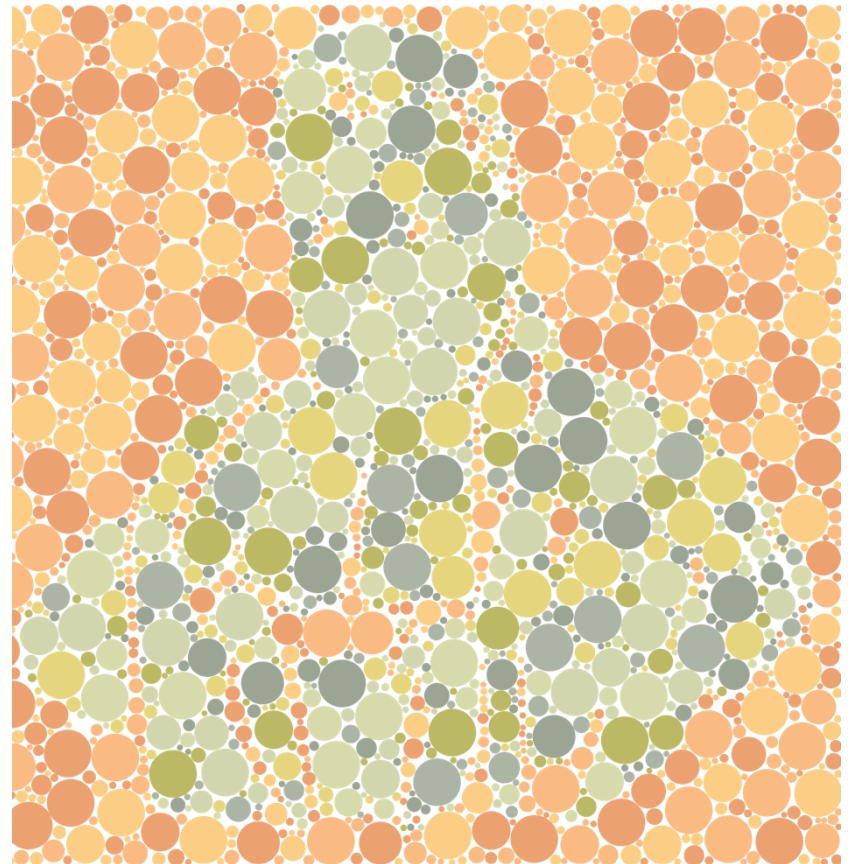
If $P(F) > 0$ then

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Example: Color Blindness

Color blindness is caused by a genetic mutation and results in people not being able to distinguish certain colors.

- The probability of being a male and having color blindness is 0.05
- The probability of being a female and having color blindness is 0.005



Using conditional probability to reflect knowledge

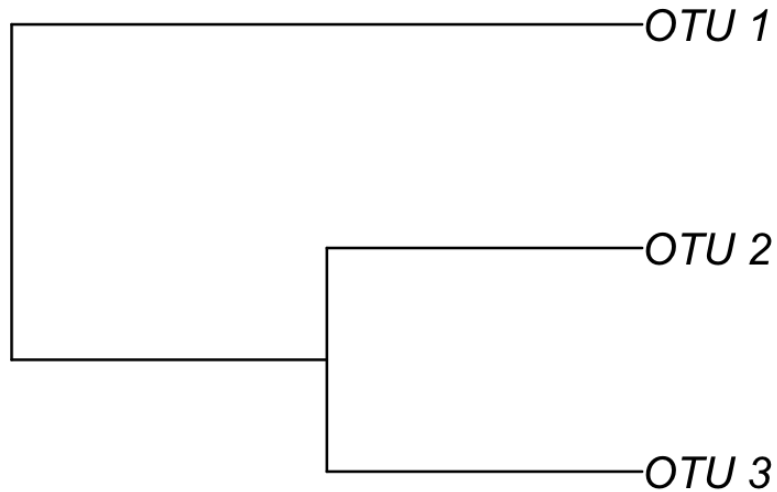
- Assuming that there is an equal number of males and females in the population, what is the probability of an individual being color blind, given that the individual is a male?
- Assuming that there is an equal number of males and females in the population, what is the probability of an individual being color blind, given that the individual is a female?



Total Probability

- Assuming that there is an equal number of males and females in the population, what is the probability that a randomly chosen individual will be color blind?

Phylogenetic tree notation



Node: a point in a phylogeny where a lineage splits

- one speciation event
- common ancestor of branches that extend from it

Branches: lineages evolving through time

- the stuff between speciation events

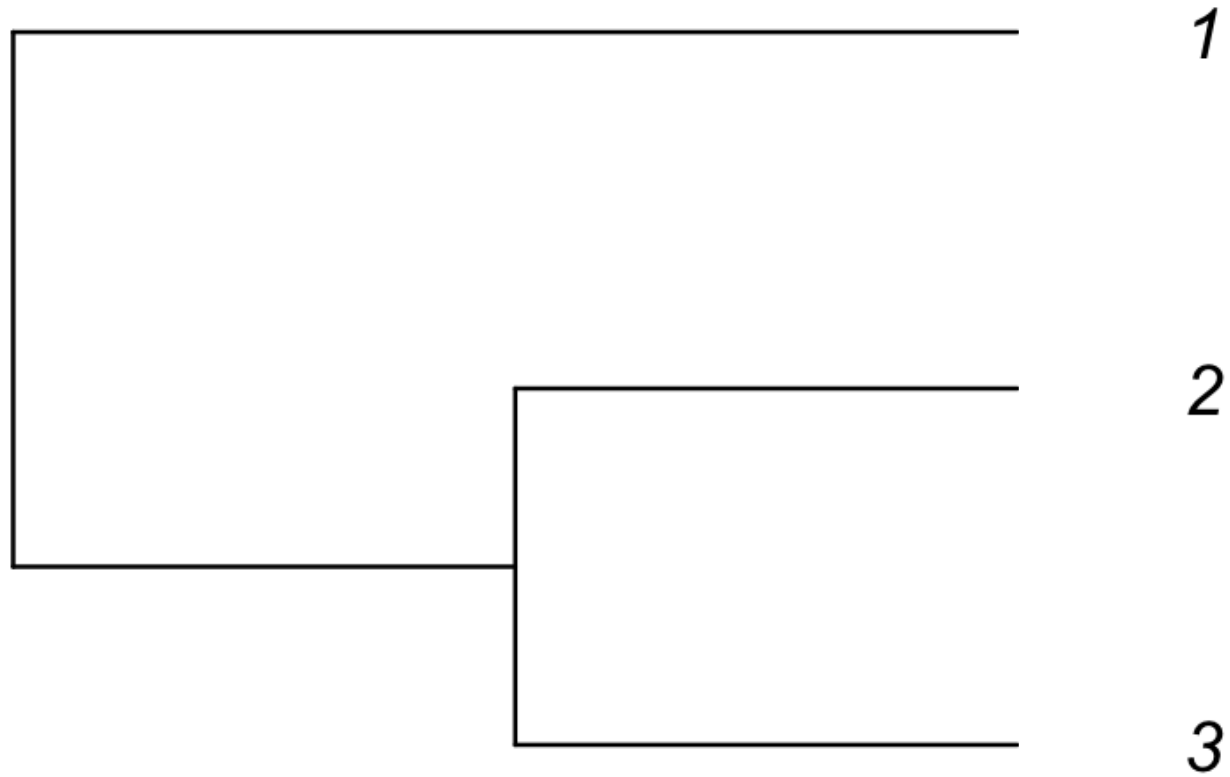
Tips: terminal ends representing species, molecules, or populations being compared

Clade: An ancestor and all its descendants

Rooted tree: includes a distantly related “outgroup” species to polarize changes and show which node is common ancestor to all lineages in ingroup

When interpreting trees, pay most attention to nodes (= common ancestors)!!

Newick format



Likelihood for a phylogenetic tree

Three essential ingredients

1. Molecular Data
2. Model for nucleotide evolution
3. A phylogenetic tree: (a) the relationships between taxa, (b) branch lengths.

Historical data: A morphological character matrix

Morphological data matrix for Carnivora

Taxa	Character-state scoring												
	#	1	2	3	4	5	6	7	8	9	10	11	12
Lemur (outgroup)	0	0	0	0	0	0	0	0	0	0	0	0	0
Cat	0	1	0	1	0	0	1	1	1	0	0	0	0
Hyena	0	1	0	1	0	0	1	0	1	0	0	0	0
Civet	0	1	0	0	0	0	0	0	1	0	0	0	0
Dog	1	0	0	0	1	0	0	0	0	0	0	0	0
Raccoon	1	0	0	0	1	0	0	0	0	0	0	0	0
Bear	1	0	0	0	1	1	0	0	0	1	0	0	0
Otter	1	0	0	0	1	0	0	0	0	1	0	0	0
Seal	1	0	1	0	1	1	0	0	0	1	1	1	0
Walrus	1	0	1	0	1	1	0	0	0	1	1	1	0
Sea lion	1	0	1	0	1	1	0	0	0	1	0	0	0

Lots of these...

1	1	1	1	0	0	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	0
0	0	0	0	1	1	1

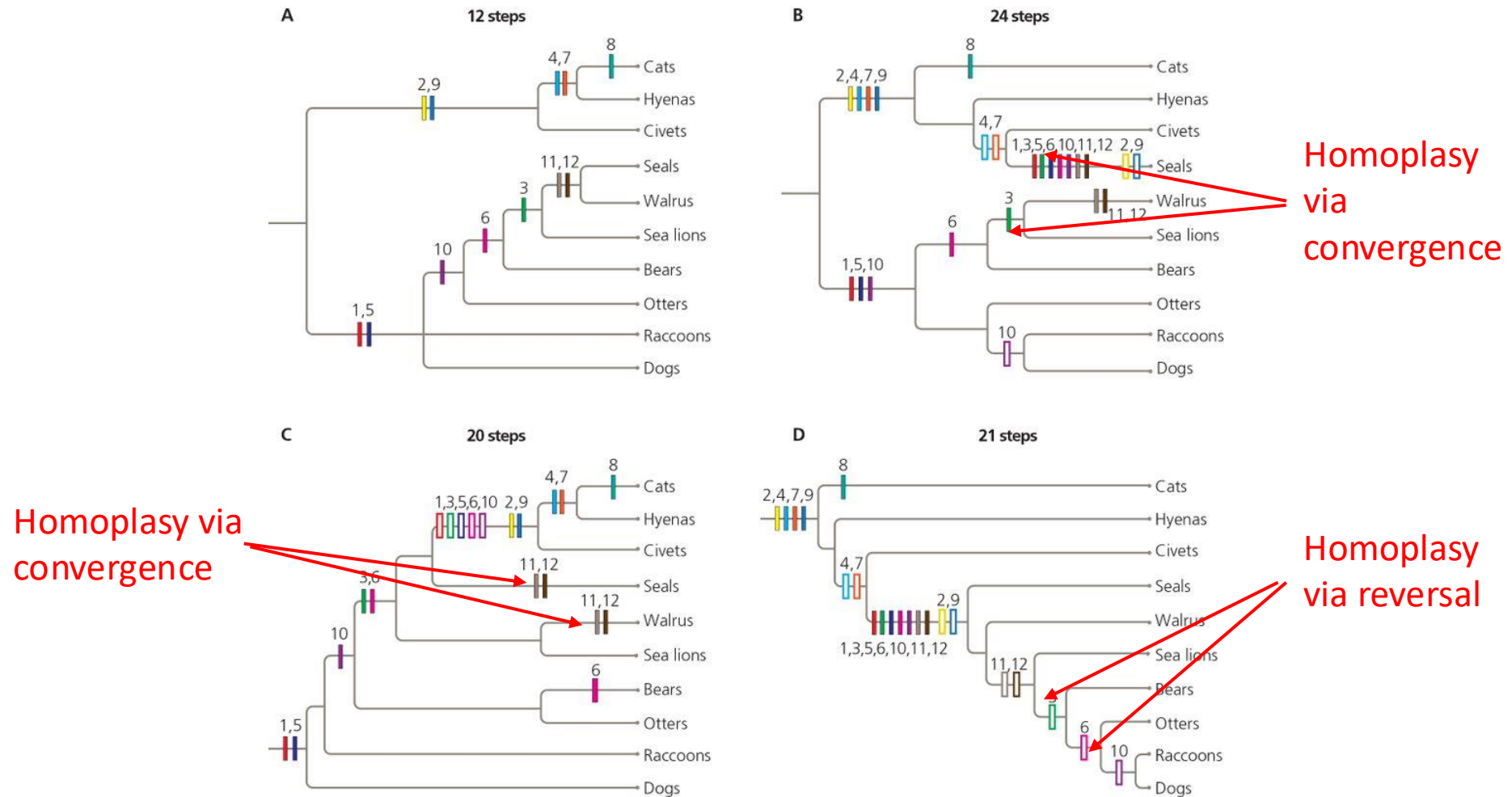
All character states in outgroup are ancestral

This does NOT mean that the outgroup lacks derived traits!

Not informative because they are not synapomorphies in ingroup.

Outgroups help us identify **shared derived states (synapomorphies)**

How to choose a tree that best explains the data



Bars = synapomorphies (shared, derived traits)
Open bars = reversion to ancestral-like state

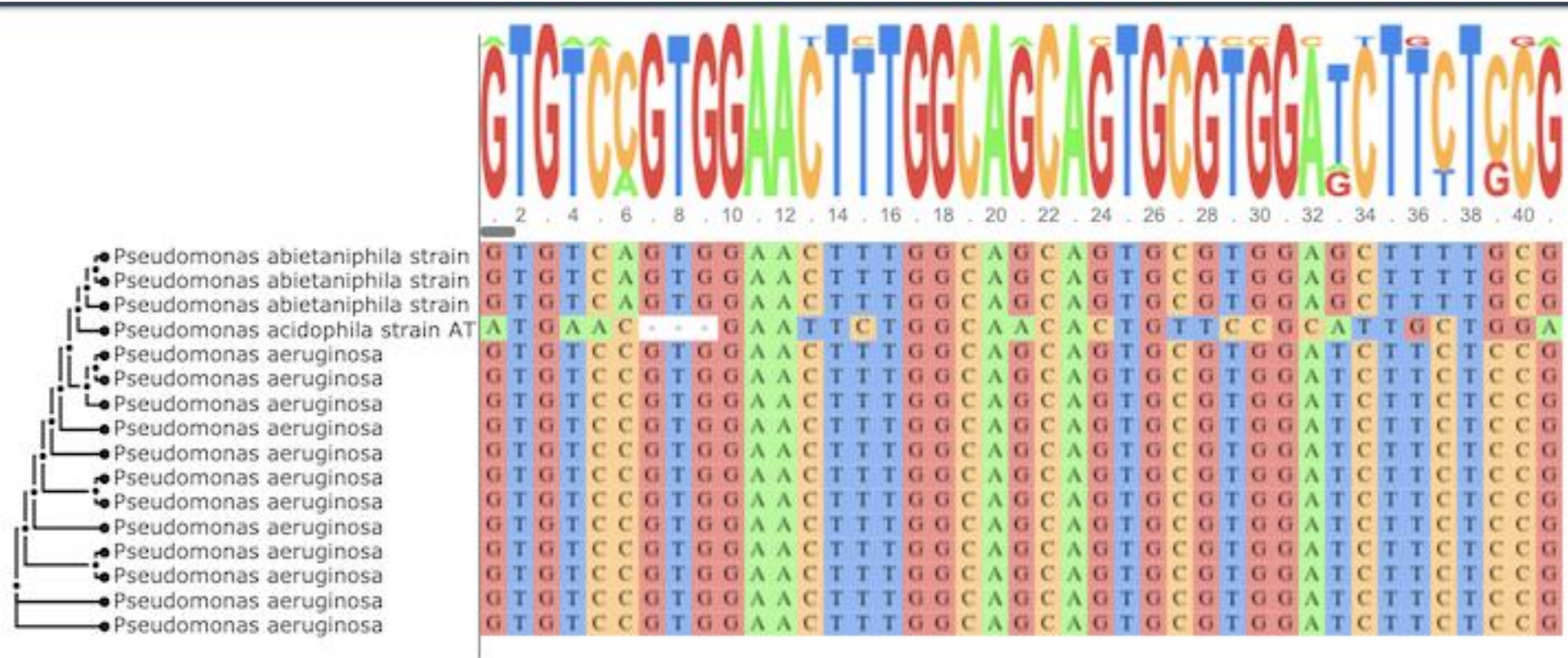
Parsimony analysis in practice...

TABLE 9.1 The Huge Number of Possible Tree Topologies

# of Taxa	# Unrooted trees	# Rooted trees
1	1	1
2	1	1
3	1	3
4	3	15
5	15	105
6	105	945
7	945	10,395
8	10,395	135,135
9	135,135	2,027,025
10	2,027,025	34,459,425
11	34,459,425	654,729,075
12	654,729,075	13,749,310,575
13	13,749,310,575	316,234,143,225

Even with computers, trees cannot be exhaustively searched for most analyses so these programs employ algorithms to efficiently search “tree space”

1. Molecular Data

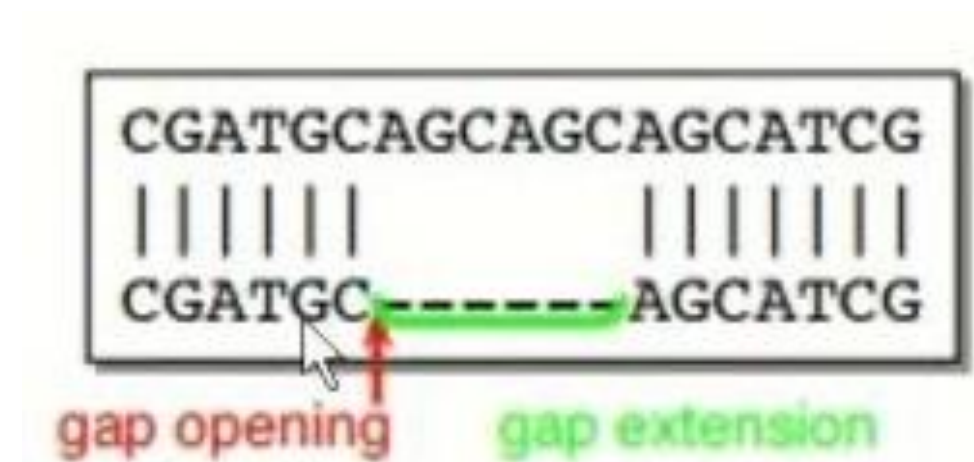
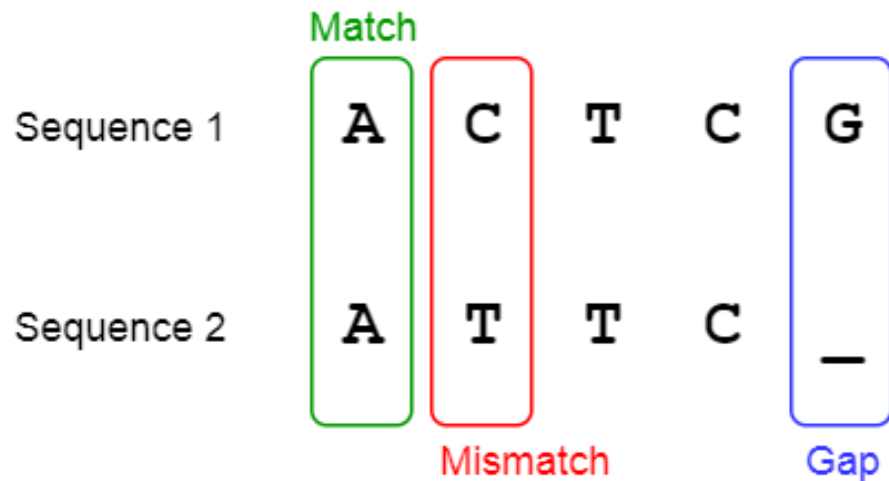


Multiple sequence alignment

- Multiple sequence alignment (MSA) is important for phylogenetic estimation or model-based inference of evolutionary processes
- The goal of MSA is to introduce gaps into sequences so that columns of an aligned matrix contain character states that are homologous
- Homology cannot be directly observed but can be inferred

Inferring Homology

- Placing gaps in a sequence is penalized too
- Introducing a new gap usually has a higher cost than extending an existing gap



Generic Alignment Scoring Parameters

match = +5
mismatch = -3
gap open = -5
gap extension = -2

Example 1

Alignment 1

AGTTCCCTG
AGTTA--TG

Score

20

In example 1, the first alignment has a higher score for minimizing gap openings

Alignment 2

AGTTCCCTG
AGTT-A-TG

17

Example 2

Alignment 1

AGTTCCACTG
AGTTA---TG

18

In example 2, the score improvement from an extra match outweighs the cost of an extra gap opening

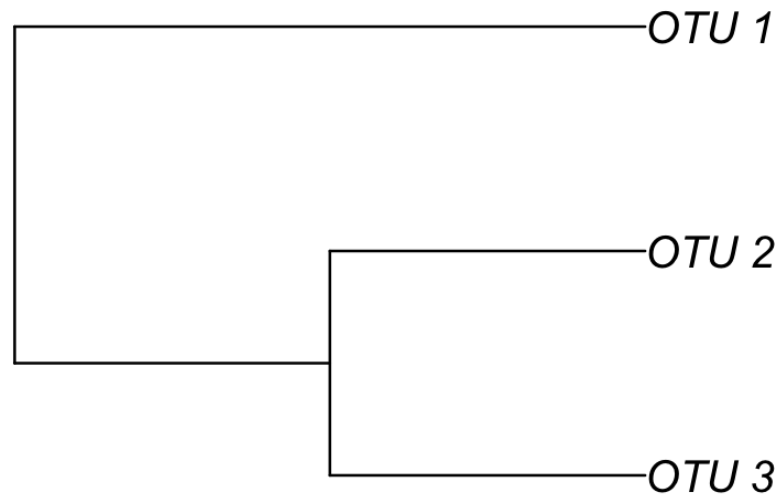
Alignment 2

AGTTCCACTG
AGTT--A-TG

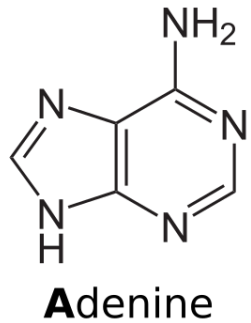
23

Likelihood in a phylogenetic tree

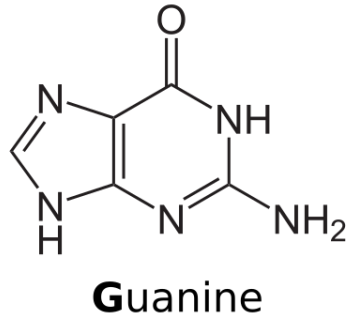
Taxon	Site 1	Site 2	Site 3	Site 4
OTU 1	G	G	G	G
OTU 2	G	A	A	G
OTU 3	A	A	G	A



2. Model for nucleotide substitutions Probability Matrix (Conditional Probability)



Purines



Assumptions

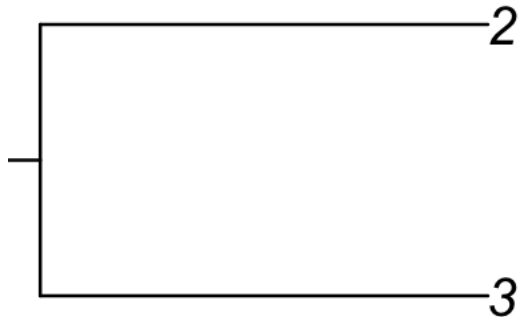
(1)

Taxon	Site 1
OTU 1	G
OTU 2	G
OTU 3	A

Site 1 $\perp$ Site 2

Taxon	Site 2
OTU 1	G
OTU 2	A
OTU 3	A

(2)

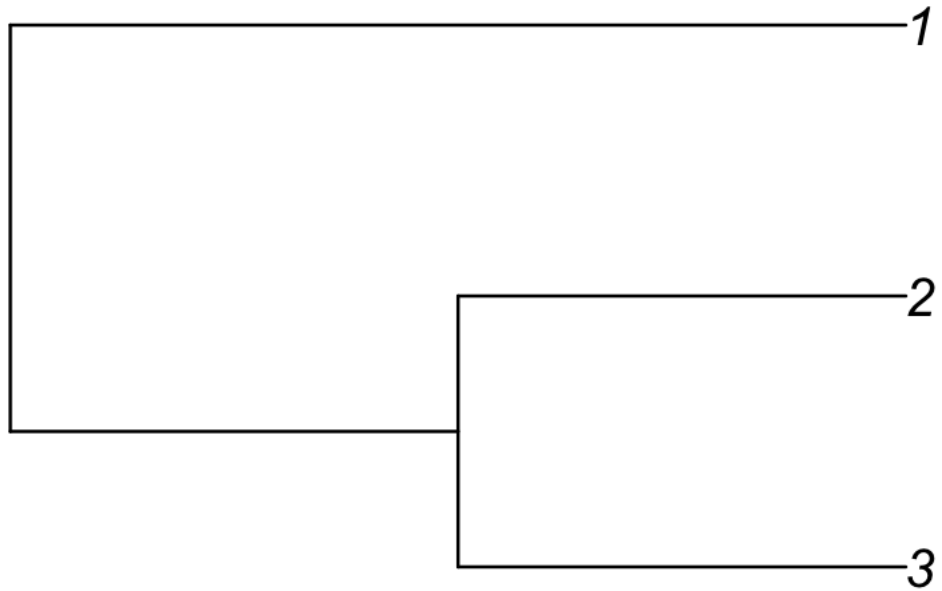


Conditional independence

When conditioning over the most recent common ancestor two lineages are independent

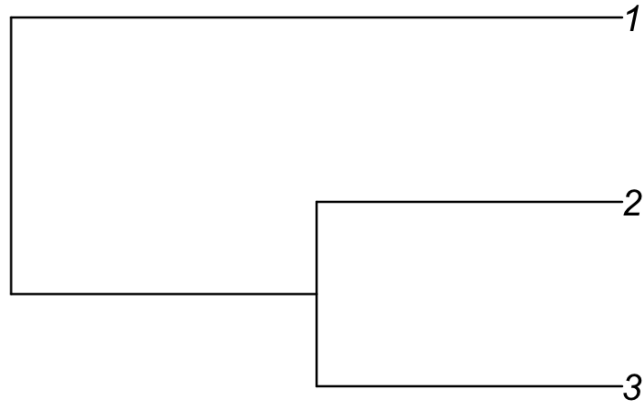
One probable story for Site 1

Taxon	Site 1
OTU 1	G
OTU 2	G
OTU 3	A

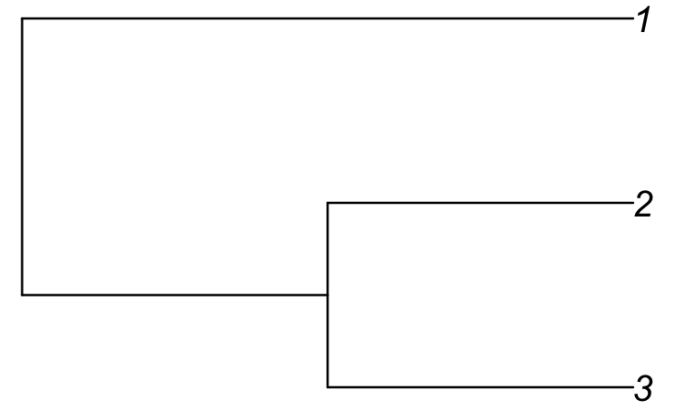


All the probable stories for Site 1

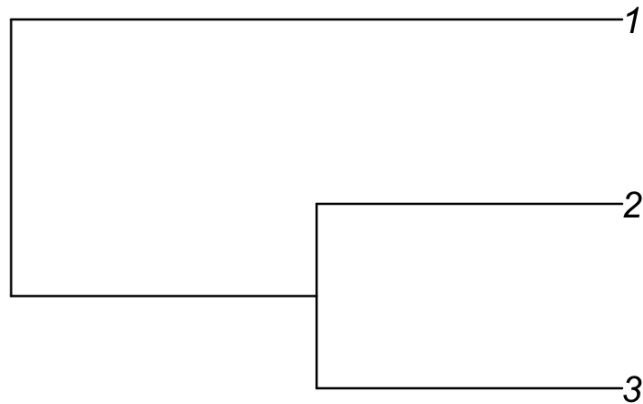
$$p_{AA}^{v4} \times p_{AG}^{v3} \times p_{AA}^{v2} \times p_{AG}^{v1} \times p_0(A)$$



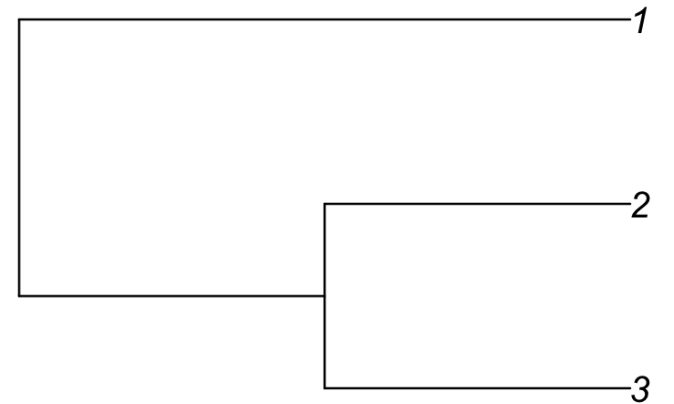
$$p_{GA}^{v4} \times p_{GG}^{v3} \times p_{AG}^{v2} \times p_{AG}^{v1} \times p_0(G)$$



$$p_{AA}^{v4} \times p_{AG}^{v3} \times p_{GA}^{v2} \times p_{GG}^{v1} \times p_0(A)$$



$$p_{GA}^{v4} \times p_{GG}^{v3} \times p_{GG}^{v2} \times p_{GG}^{v1} \times p_0(G)$$

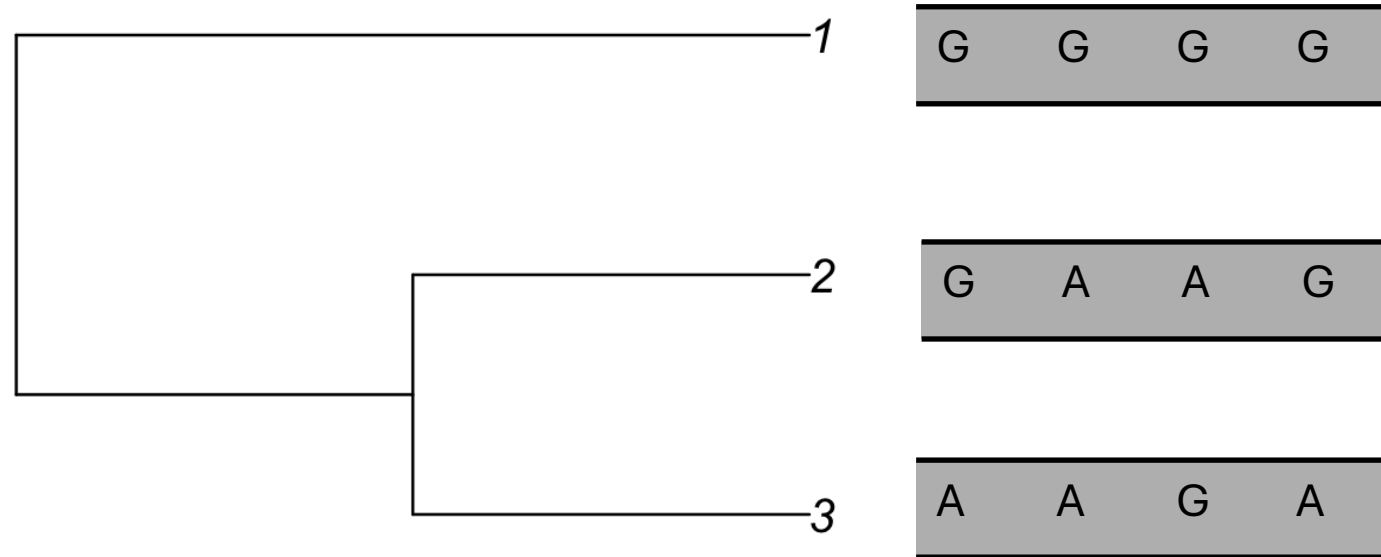


Probability for a single site given one tree and model

$$P(\text{Site } 1 | \psi_1, \text{Model}) = \sum_{j=A}^G \left(\sum_{i=A}^G (p_{iA}^{v4} \times p_{iG}^{v3}) \times p_{ij}^{v2} \right) \times p_{jG}^{v1} \times p_0(j)$$

Likelihood for all sites and one tree under one model

$$\mathcal{L}(\psi_1; \text{Data}, \text{Model (P)}) = \prod_{k=1}^4 P(\text{Site } k | \psi_1, \text{Model (P)})$$

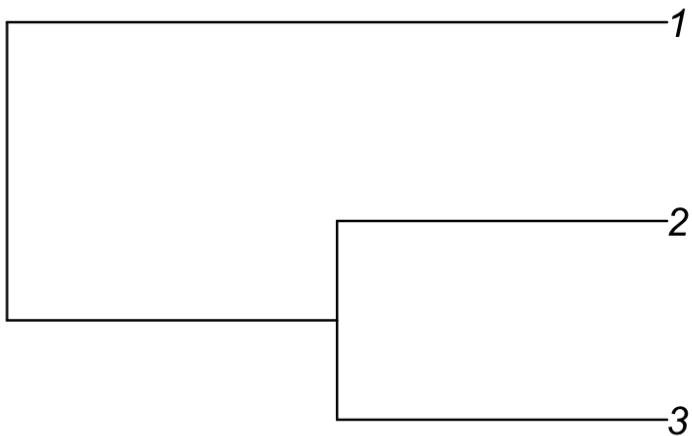


Maximum likelihood phylogenetic tree (MLE)

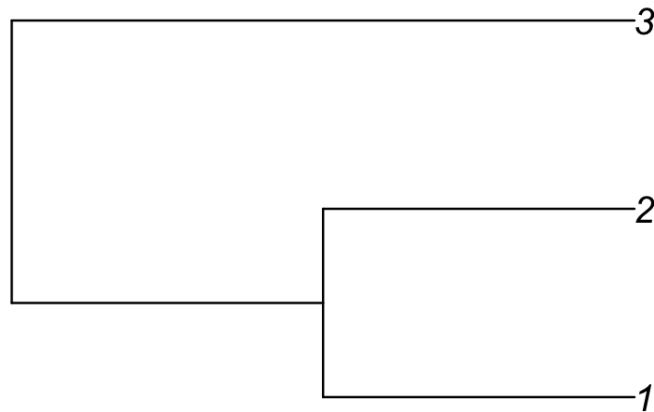
Which tree maximizes the likelihood function?

$$\max_{k=1,2,3} (\mathcal{L}(\text{Data}; \text{Model (P)}, \psi_k))$$

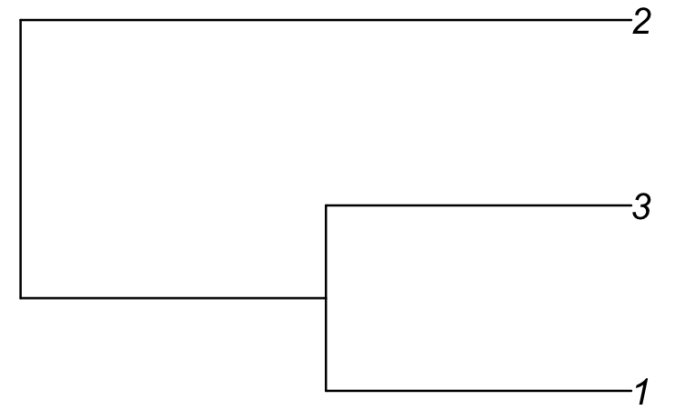
$\mathcal{L}(\text{Data}; \text{Model (P)}, \psi_1)$



$\mathcal{L}(\text{Data}; \text{Model (P)}, \psi_2)$



$\mathcal{L}(\text{Data}; \text{Model (P)}, \psi_3)$



What is changing in this likelihood?

Inferring a phylogeny using the likelihood function



Model for molecular evolution



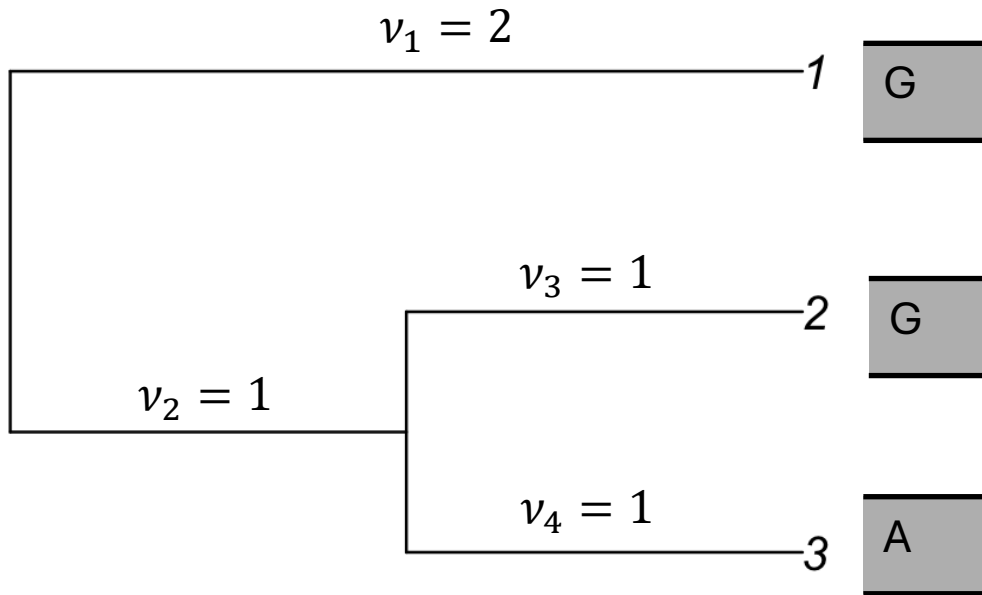
A mechanism to propose trees



Assumption about branch
lengths

Exercise:

Calculate the probability of this site given this phylogenetic tree when the model changes by branch length.



$$P(1) = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$$

$$P(2) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$